

YUFENG LIU

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Education

Carleton University

Ottawa, ON

Bachelor in Computer Science, AI/ML Stream, 3.71/4 GPA

Sept 2022 - Apr 2027

- Third Year Standing | Available for 3 months beginning June 2026 | Cansbridge Scholar | Study Abroad to Japan 2026

Technical Skills

Languages: Python, C/C++, Typescript/JavaScript, SQL/PostgreSQL

Developer Tools: Git, Terminal, Unix/Linux, VSCode, Figma, CI/CD Pipeline, ORMs, JIRA, Agile, BitBucket

Technologies/Frameworks: NumPy, pandas, React, Node.js, Selenium, Express.js, Prisma, PyTorch, TensorFlow, Keras

Work Experience

Planetary Robotics Intern | *Python, Django, React*

Montreal, QC

Canadian Space Agency

Sept. 2025 - Dec. 2025

- Developed a **Gantt chart interface using svar-gantt** (~1500 lines of code) that visualizes activity durations, rover traverse times, and science operations across multiple regions of interest, **consolidating strategic and tactical planning into a single view** with drag-to-reschedule functionality to reorder pathfinding sequences
- Implemented throughpoints feature **end-to-end** by creating database models, backend API controllers for route calculation, and frontend UI with toolbar integration and interactive map placement, enabling scientists to force rover routing through specific coordinates to maintain line-of-sight communication with the lander

Technical Project Manager Intern | *Python, Next.js, pandas, React*

Ottawa, ON

Blackberry QNX

May 2025 - Aug. 2025

- **Developed first-ever dependency visualization tool for QNX's 4400 binaries across 3 versions (QNX700/710/800)**, enabling internal developers to instantly identify impact of code changes
- Reduced dependency discovery time by 96% and **optimized data scraping from 40 hours to 20 minutes (120x faster)** through multithreading and batch processing

Software Engineer Intern | *Python, pandas, NumPy, HuggingFace, BitBucket, JIRA*

Ottawa, ON

Ciena

Sept 2024 - Dec 2024

- Pioneered and implemented Ciena's **first-ever locally-hosted syslog analysis AI system** using **Hugging Face's Transformers API** to fine-tune a Large Language Model (*led-large-16384*), achieving a **96% performance improvement** over the base model, enabling automated interpretation of thousands of technical error messages
- Architected a robust data pipeline for AI training by developing a Python-based **templating system** to reverse engineer syslog patterns, enabling scalable synthetic data generation and dramatically expanding the training dataset

Volunteer Software Developer | *HTML/CSS, Javascript, Prisma, Node.js, PostgreSQL, Jest*

Ottawa, ON

Carleton Blueprint

May 2024 - Dec 2024

- Engineered a sophisticated student-mentor matching system using **Prisma ORM** and **PostgreSQL**, optimizing academic support by efficiently pairing struggling students with appropriate mentors
- Implemented comprehensive unit testing suite with **Jest**, ensuring robust data integrity and accurate instantiation in the PostgreSQL database via Prisma ORM, significantly enhancing system reliability and performance

Personal Projects

Artificial Intelligence / ML | *Python, Selenium, pandas, NumPy, Tensorflow, PyTorch*

- Engineered a **web scraper leveraging Large Language Model** to extract data from websites, implementing advanced preprocessing techniques that optimized tokenization efficiency and reduced token costs by approximately **10x**
- Implemented a **Linear Regression** model to analyze housing data, incorporating **Z-score normalization** and **gradient descent** algorithms; achieving an accurate model to predict local housing data off of 4 inputs
- Architected and constructed a custom **Neural Network** and **Large Language Model** from the ground up using **TensorFlow and PyTorch**, in order to fully understand Deep Learning principles including backpropagation, activation functions, and model optimization techniques

SPUT - Simple Programmable Universal Terrier ([Link](#)) | *Python, Raspberry Pi, ROS2, Fusion360*

- Architecting an **autonomous robotic dog from scratch** including 3D mechanical design in Fusion360, servo control systems, computer vision integration with OpenCV, and complex inverse kinematics algorithms for coordinated multi-limb movement and obstacle navigation